

Quantum Field Theory – Homework 2 (Solutions)

Name: _____ Due Date: 2025.11.03

Problem 1. Probability Interpretation and the Klein–Gordon Equation (20 points)

In non-relativistic quantum mechanics, the Schrödinger equation

$$i \frac{\partial}{\partial t} \psi(\vec{x}, t) = -\frac{1}{2m} \nabla^2 \psi(\vec{x}, t)$$

admits a conserved quantity interpreted as probability. Multiplying by ψ^* and subtracting the complex conjugate gives

$$\frac{\partial}{\partial t} |\psi|^2 + \nabla \cdot \left[\frac{i}{2m} (\psi \nabla \psi^* - \psi^* \nabla \psi) \right] = 0.$$

Hence we can identify

$$\rho = |\psi|^2, \quad \vec{J} = \frac{i}{2m} (\psi \nabla \psi^* - \psi^* \nabla \psi),$$

which satisfy the continuity equation $\partial_t \rho + \nabla \cdot \vec{J} = 0$, so that ρ is real, non-negative, and represents the probability density.

Now consider the relativistic Klein–Gordon equation for a single relativistic particle

$$\partial_\mu \partial^\mu \psi(\vec{x}, t) - m^2 \psi(\vec{x}, t) = 0.$$

- (a) Derive the conserved current associated with this equation by combining it with its complex conjugate. Show that it can be written as

$$\partial_\mu j^\mu = 0, \quad j^\mu = i(\psi^* \partial^\mu \psi - \psi \partial^\mu \psi^*).$$

Identify the time component j^0 and spatial component $\vec{j}$.

Starting from the Klein–Gordon equation

$$(\partial_\mu \partial^\mu - m^2) \psi = 0,$$

and its complex conjugate

$$(\partial_\mu \partial^\mu - m^2) \psi^* = 0,$$

multiply the first by ψ^* and the second by ψ and subtract:

$$\psi^*(\partial_\mu\partial^\mu\psi) - \psi(\partial_\mu\partial^\mu\psi^*) = 0.$$

Rewriting gives

$$\partial_\mu(\psi^*\partial^\mu\psi - \psi\partial^\mu\psi^*) = 0.$$

Hence the conserved current is

$$\boxed{j^\mu = i(\psi^*\partial^\mu\psi - \psi\partial^\mu\psi^*) \quad \text{with} \quad \partial_\mu j^\mu = 0.}$$

The time component and spatial part are

$$j^0 = i(\psi^*\dot{\psi} - \psi\dot{\psi}^*), \quad \vec{j} = i(\psi^*\nabla\psi - \psi\nabla\psi^*).$$

- (b) Examine the expression for j^0 and discuss its sign. Why can j^0 take negative values even for well-behaved wave packets? Explain why it therefore cannot be interpreted as a probability density.

The quantity $j^0 = i(\psi^*\dot{\psi} - \psi\dot{\psi}^*)$ is not positive-definite. For plane-wave solutions $\psi \sim e^{-iEt+i\vec{k}\cdot\vec{x}}$, we have

$$j^0 = 2E|\psi|^2.$$

For positive-frequency modes $E > 0$, $j^0 > 0$, but for negative-frequency solutions $E < 0$, $j^0 < 0$. Hence, j^0 can take either sign and cannot be interpreted as a probability density.

- (c) Briefly explain the physical implication of this result. Why does the failure of a positive-definite probability density indicate that the Klein-Gordon equation cannot consistently describe a single relativistic particle, and motivates the need for a field-theoretic description?

Because j^0 is not positive definite, the Klein-Gordon equation cannot describe the probability density of a single relativistic particle. Instead, ψ must be interpreted as a **field**, not a wavefunction, whose excitations correspond to particles and antiparticles. This failure motivates the transition from relativistic quantum mechanics to **quantum field theory**.

Problem 2. Noethers Theorem and Energy-Momentum Conservation (20 points)

Consider the real scalar field theory

$$S[\phi] = \int d^4x \left[-\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi) \right].$$

- (a) Show that the action is invariant under spacetime translations $x^\mu \rightarrow x^\mu + a^\mu$.

Under an infinitesimal translation $x^\mu \rightarrow x'^\mu = x^\mu + a^\mu$, the field transforms as

$$\delta\phi = -a^\nu \partial_\nu \phi.$$

The Lagrangian density changes by a total derivative:

$$\delta\mathcal{L} = -a^\nu \partial_\nu \mathcal{L}.$$

Since the action changes only by a boundary term, $\delta S = \int d^4x \partial_\nu (a^\nu \mathcal{L}) = 0$, the action is invariant under translations.

- (b) Using Noethers theorem, derive the conserved current associated with this symmetry and identify the corresponding conserved quantity.

From Noethers theorem, the conserved current associated with translation invariance is

$$j_{(\nu)}^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\nu \phi - \delta^\mu_\nu \mathcal{L}.$$

This is identified as the **energy-momentum tensor**:

$$\boxed{T^\mu_\nu = \partial^\mu \phi \partial_\nu \phi - \delta^\mu_\nu \mathcal{L}, \quad \partial_\mu T^\mu_\nu = 0.}$$

- (c) Write down the components of the energy-momentum tensor $T^{\mu\nu}$ for this theory.

Writing $\mathcal{L} = -\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi)$, the components of $T^{\mu\nu}$ are

$$\boxed{T^{\mu\nu} = \partial^\mu \phi \partial^\nu \phi - g^{\mu\nu} \left[\frac{1}{2} (\partial_\alpha \phi)(\partial^\alpha \phi) + V(\phi) \right].}$$

For example, T^{00} is the energy density and T^{0i} the momentum density.

Problem 3. Lorentz Invariance of the Scalar Field Action (15 points)

Show explicitly that the scalar field action

$$S = \int d^4x \left[-\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 \right]$$

is invariant under an infinitesimal Lorentz transformation

$$x^\mu \rightarrow x'^\mu = x^\mu + \Lambda^\mu{}_\nu x^\nu, \quad \phi(x) \rightarrow \phi'(x') = \phi(x),$$

where $\Lambda_{\mu\nu} = -\Lambda_{\nu\mu}$.

Under an infinitesimal Lorentz transformation

$$x^\mu \rightarrow x'^\mu = x^\mu + \Lambda^\mu{}_\nu x^\nu, \quad \phi'(x') = \phi(x),$$

the derivatives transform as

$$\partial_\mu \phi(x) \rightarrow \partial'_\mu \phi'(x') = (\delta_\mu{}^\nu - \Lambda_\mu{}^\nu) \partial_\nu \phi(x).$$

Substitute into the action

$$S = \int d^4x \left[-\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 \right].$$

Because $\Lambda_{\mu\nu} = -\Lambda_{\nu\mu}$, the quadratic term $\partial_\mu \phi \partial^\mu \phi$ is invariant up to a total derivative, and d^4x is Lorentz invariant. Thus $\delta S = 0$ and the action is Lorentz invariant.

Then, use Noether's theorem to find the expression for the conserved angular-momentum current $M^{\lambda\mu\nu}$.

The Noether current associated with Lorentz invariance is the **angular-momentum tensor**

$$M^{\lambda\mu\nu} = x^\mu T^{\lambda\nu} - x^\nu T^{\lambda\mu}.$$

It encodes both orbital and spin angular momentum. For a real scalar field (no spin), the second term vanishes, leaving only the orbital part above.

Problem 4. From Discrete Systems to Field Theory (15 points)

Starting from the Lagrangian of a one-dimensional chain of coupled oscillators

$$L = \sum_n \left[\frac{1}{2} \mu \dot{\eta}_n^2 - \frac{\lambda}{2} (\eta_{n+1} - \eta_n)^2 - \frac{1}{2} \sigma \eta_n^2 \right],$$

derive its continuum limit as the lattice spacing $a \rightarrow 0$, obtaining a field theory in terms of a continuous field $\eta(x, t)$. Identify the effective mass m and propagation speed v , and discuss under what condition this theory becomes relativistic.

Start from

$$L = \sum_n \left[\frac{1}{2} \mu \dot{\eta}_n^2 - \frac{\lambda}{2} (\eta_{n+1} - \eta_n)^2 - \frac{1}{2} \sigma \eta_n^2 \right].$$

Let $x = na$ and $\eta_n(t) \rightarrow \eta(x, t)$. For small lattice spacing a ,

$$\eta_{n+1} - \eta_n \approx a \partial_x \eta,$$

and the sum becomes an integral

$$\sum_n \rightarrow \frac{1}{a} \int dx.$$

Then

$$L = \int dx \left[\frac{1}{2} \mu (\partial_t \eta)^2 - \frac{1}{2} \lambda a^2 (\partial_x \eta)^2 - \frac{1}{2} \sigma \eta^2 \right].$$

Define the continuum field theory Lagrangian density

$$\mathcal{L} = \frac{1}{2} \mu (\partial_t \eta)^2 - \frac{1}{2} \lambda a^2 (\partial_x \eta)^2 - \frac{1}{2} \sigma \eta^2.$$

We can identify

$$v^2 = \frac{\lambda a^2}{\mu}, \quad m^2 = \frac{\sigma}{\mu}.$$

Hence, in the continuum limit,

$$\boxed{\mathcal{L} = \frac{1}{2} (\partial_t \eta)^2 - \frac{1}{2} v^2 (\partial_x \eta)^2 - \frac{1}{2} m^2 \eta^2.}$$

The equation of motion is

$$\partial_t^2 \eta - v^2 \partial_x^2 \eta + m^2 \eta = 0.$$

The theory becomes **relativistic** when $v = 1$, so that time and space derivatives enter symmetrically.